

Experiment-6

Aim: Write a program to render HTML to a web page.

Description:

- ReactJS is a JavaScript library used to create interactive user interfaces.
- HTML content is rendered to the webpage using ReactDOM.render().
- A class component is created by extending React.Component.
- The component returns JSX (HTML-like syntax).
- JSX allows writing HTML inside JavaScript.
- The same component is reused multiple times on the page.
- React renders content inside a selected DOM element (#container).
- CSS is used to style the rendered output.
- React uses the Virtual DOM for efficient updating of the UI. **Comp1.html**

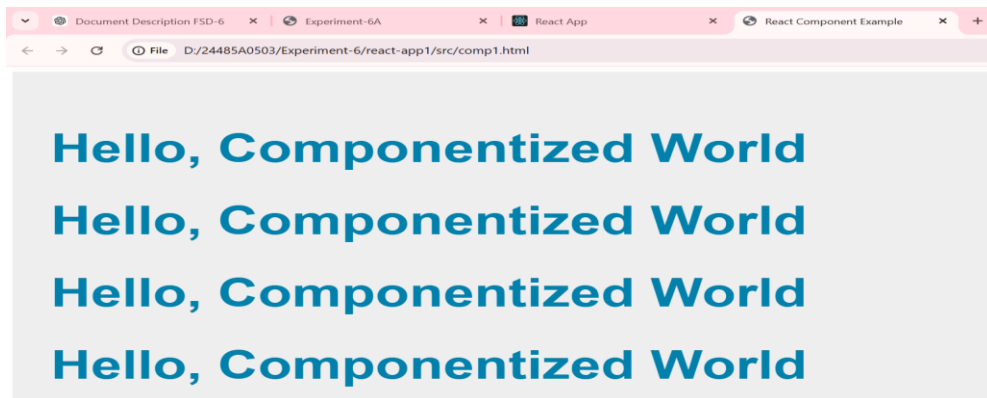
Program:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="UTF-8" />
    <title>React Component Example</title>
    <script src="https://unpkg.com/react@18/umd/react.development.js" crossorigin></script>
    <script src="https://unpkg.com/react-dom@18/umd/react-dom.development.js"
crossorigin></script>
    <script src="https://unpkg.com/@babel/standalone/babel.min.js"></script>
    <style>
      #container {
        padding: 50px;
        background-color: #EEE;
      }
      #container h1 {
        font-size: 72px;
        font-family: sans-serif;
        color: #0080AB;
      }
    </style>
  </head>
  <body>
    <div id="container"></div>
    <script type="text/babel">
      const container = document.querySelector("#container");
```

```

const root = ReactDOM.createRoot(container);
class HelloWorld extends React.Component {
  render() {
    return <h1>Hello, Componentized World</h1>;
  }
}
root.render(
  <div>
    <HelloWorld />
    <HelloWorld />
    <HelloWorld />
    <HelloWorld />
  </div>
);
</script>
</body>
</html>

```

Output:

b) **Aim:** Write a program for writing markup with JSX.

Description:

- JSX stands for JavaScript XML and allows writing HTML-like markup inside JavaScript.
- JSX makes UI code easier to read and write.
- Curly braces {} in JSX are used to insert dynamic values.
- Props (properties) are used to pass data from parent to component.
- this.props is used in class components to access passed data.
- Components can display different outputs based on the props received.
- React renders multiple components with different data using JSX.
- JSX is compiled into regular JavaScript using Babel.
- This demonstrates dynamic content rendering using JSX and props. **Comp2.html**

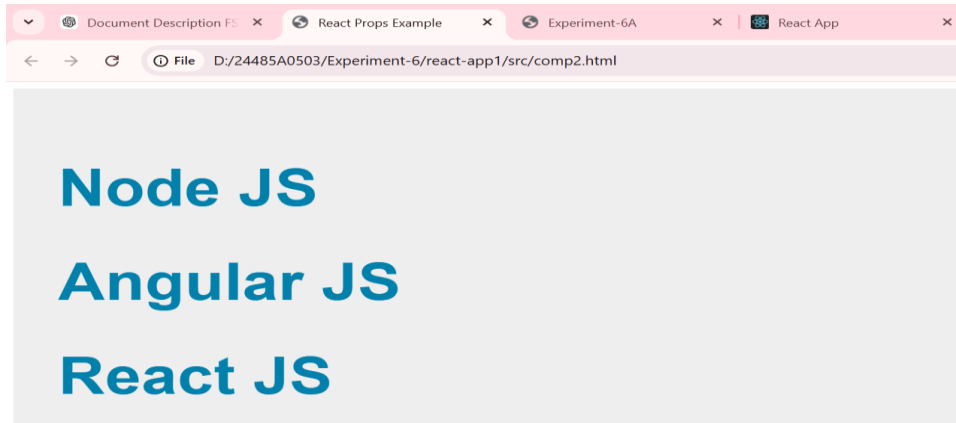
Program:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="UTF-8" />
    <title>React Props Example</title>
    <script src="https://unpkg.com/react@18/umd/react.development.js" crossorigin></script>
    <script src="https://unpkg.com/react-dom@18/umd/react-dom.development.js"
crossorigin></script>
    <script src="https://unpkg.com/@babel/standalone/babel.min.js"></script>
    <style>
      #container {
        padding: 50px;
        background-color: #EEE;
      }
      #container h1 {
        font-size: 72px;
        font-family: sans-serif;
        color: #0080AB;
      }
    </style>
  </head>
  <body>
    <div id="container"></div>
    <script type="text/babel">
      const destination = document.querySelector("#container");
      const root = ReactDOM.createRoot(destination);
      class HelloWorld extends React.Component {
        render() {
          return <h1>{this.props.greetTarget}</h1>;
        }
      }
      root.render(
        <div>
          <HelloWorld greetTarget="Node JS" />
          <HelloWorld greetTarget="Angular JS" />
          <HelloWorld greetTarget="React JS" />
        </div>
      );
    </script>
  </body>
</html>
```

```

</script>
</body>
</html>

```

Output:

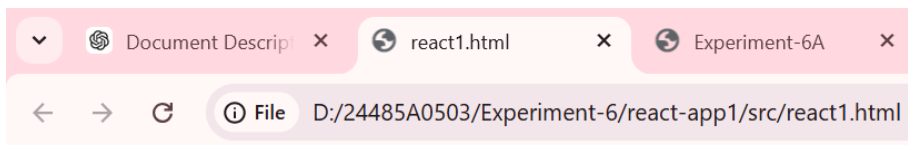
C) **Aim:** Write a program for creating and nesting components (function and class).

Program:

```

<html>
  <head>
    <script src="https://unpkg.com/react@18/umd/react.development.js" crossorigin></script>
    <script src="https://unpkg.com/react-dom@18/umd/react-dom.development.js"
crossorigin></script>
    <script src="https://unpkg.com/@babel/standalone/babel.min.js"></script>
  </head>
  <body >
    <script type="text/babel">
      ReactDOM.render(<h1> Welcome To ReactJS Program</h1>,document.body);
    </script>
  </body>
</html>

```

Output:

Welcome To ReactJS Program

ii)react2.html Program:

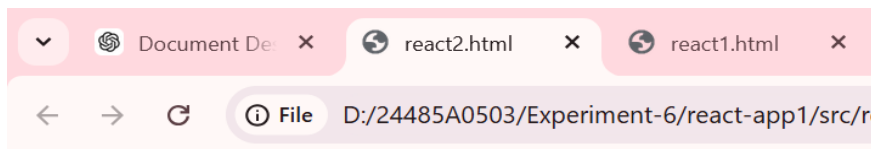
Program:

```

<html>
  <head>
    <script src="https://unpkg.com/react@18/umd/react.development.js" crossorigin></script>
    <script src="https://unpkg.com/react-dom@18/umd/react-dom.development.js"
crossorigin></script>
    <script src="https://unpkg.com/@babel/standalone/babel.min.js"></script>
  </head>
  <body >
<div id="container"></div>

    <script type="text/babel">
      ReactDOM.render(
        <h1> Welcome To ReactJS Program</h1>,
        document.querySelector("#container"));
    </script>
  </body>
</html>

```

Output:

Welcome To ReactJS Program

iii) react3.html Program:**Program:**

```

<html>
  <head>
    <script src="https://unpkg.com/react@18/umd/react.development.js" crossorigin></script>
    <script src="https://unpkg.com/react-dom@18/umd/react-dom.development.js"
crossorigin></script>
    <script src="https://unpkg.com/@babel/standalone/babel.min.js"></script>
  </head>
  <body >
<div id="container"></div>

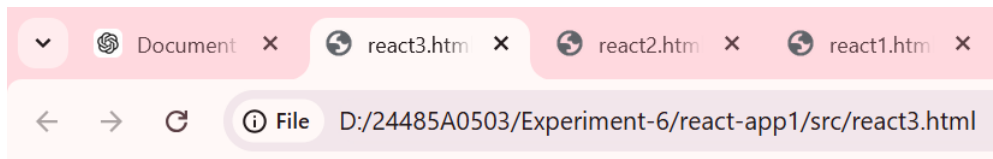
```

```

<script type="text/babel">
var destination=document.querySelector("#container");
  ReactDOM.render(
    <h1> Welcome To ReactJS Program</h1>,
    destination);
</script>

</body>
</html>

```

Output:

Welcome To ReactJS Program

iv)react4.html**Program:**

```

<html>
<head>
  <meta charset="UTF-8" />
  <title>React Basic Example</title>
  <script src="https://unpkg.com/react@18/umd/react.development.js" crossorigin></script>
  <script src="https://unpkg.com/react-dom@18/umd/react-dom.development.js"
crossorigin></script>
  <script src="https://unpkg.com/@babel/standalone/babel.min.js"></script>
  <style>
    #container {
      padding: 50px;background-color: #EEE;
    }
    #container h1 {
      font-size: 72px;font-family: sans-serif;color: #0080AB;
    }
  </style>
</head>
<body>

```

```
<div id="container"></div>
<script type="text/babel">
  const container = document.querySelector("#container");
  const root = ReactDOM.createRoot(container);
  root.render(
    <h1>Welcome To ReactJS Program</h1>
  );
</script>
</body>
</html>
```

Output: